[scale=1.5] ver-
tex=[circle,fill=black!25,minimum
size=17pt,inner sep=0pt] edge
= [draw,thick,-]

$[matrix of math nodes, nodes =$
$minimum width = 3em, minimum height = 3em, outer sep = 0pt, column sep =$
$-, row sep = -](m) A, B, C$
B, C,
C,
D, C,
E, A,
;

(a) Adjacency List for graph in figure **??**. Each line represents a vertex with the first in the list being the source and each subsequent vertex a destination of an edge.

[¿=stealth, shorten ¿=1pt,
auto, node distance=3cm,
thick, main node/.style=circle,
draw, font=]

$[matrix of math nodes, left delimiter =$
$[, right delimiter =], nodes =$
$minimum width = 3em, minimum height = 3em, outer sep = 0pt, column sep =$
$-, row sep = -](m) ABCDE$
A 0 1 1 0 0
B 0 0 1 0 0
C 0 0 0 0 0
D 0 0 1 0 0
E 1 0 0 0 0
;

(b) Adjacency Matrix for graph in figure **??** is a $|V| \times |V|$ matrix where a value of 1 indicates an edge and 0 is no edge.

[¿=stealth, shorten ¿=1pt,
auto, node distance=3cm,
thick, main node/.style=circle,
draw, font=]

$[matrix of math nodes, left delimiter =$
$[, right delimiter =], nodes =$
$minimum width = 3em, minimum height = 3em, outer sep = 0$
$-, row sep = -](M1) 11111$
1 ; [left] at (M1.west) Value
List..;
[¿=stealth, shorten ¿=1pt,
auto, node distance=3cm,
thick, main node/.style=circle,
draw, font=]

[¿=stealth, shorten ¿=1pt,
$[matrix of math nodes, left delimiter =$
$[, right delimiter =], nodes =$